

BYTE VOLTA

Country: Brazil Category : Maze

Brazil is a country known for its rich culture and growing emphasis on technology and innovation. In the Northeast region, the state of Ceará has become a rising hub for technological development, especially in the fields of robotics and automation. Ceará benefits from significant investments in digital infrastructure and a strong focus on education, which together create an environment conducive to innovation. Organizations like Farias Brito play an important role in preparing skilled professionals by combining theoretical knowledge with hands-on experience in robotics and related technologies. This educational foundation helps fuel a vibrant local community of students, engineers, and entrepreneurs who are developing advanced robotic solutions to address real-world challenges.

About us:

We are Byte Volta, the Brazilian team representing the educational institution Farias Brito. Our journey began in October 2022. The name Byte Volta is a creative play on words in our native language, Portuguese: Byte sounds like "bate" (which means "hit" or "collision") and Volta means "return" — together symbolizing both impact and resilience.



Miguel A. Santos

• Eletrical/Hardware

"I assembled the robot's mechanical components and designed the circuitry for connecting all electronic parts and designed the robot in the CAD software."



João V. G. Geiss

• Eletrical

"I assembled the robot, prototyped the PCB, managed the hardware and components, and oversaw their integration into the system."



João P. V. Silveira (C)

• Hardware/Software

"I am responsible for both hardware and software, especially in the areas of victim identification and parts of 3D modeling."



Kaleo S. Mendonça

• Software

"I oversee the overall software architecture and the exploration algorithms that drive the robot."

Awards:

2022

- 1st place LARC, Latin American Robotics Competition (Rescue Maze)
- Best Interview Award LARC (Rescue Maze)
- 4th place OBR (Rescue Line)

2023

- 20th place RoboCup (Rescue Maze)
- 4th place LARC (Rescue Maze)
- 2nd place - knowledge fair

2024

- 1st place LARC (Rescue Maze)
- 3rd place LARI Competition
- 11th place OBR (Rescue Line)

2025

- 1st place LARC (Rescue Maze)
- Participation Rescue Line
- 16 th place RoboCup (Rescue Maze)



2024

Latin American Championship

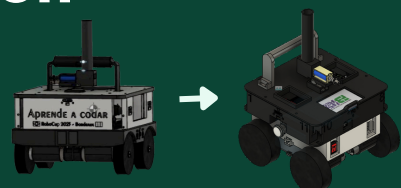


2025

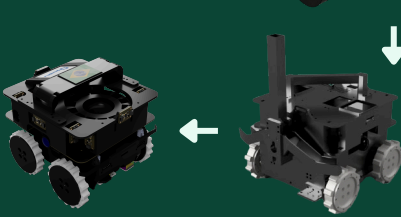
Latin American Championship

Robot Evolution

In 2023, we fixed the intermittent contact issue by implementing a custom PCB, and addressed the high current draw by replacing a Raspberry Pi with two OpenMV boards.

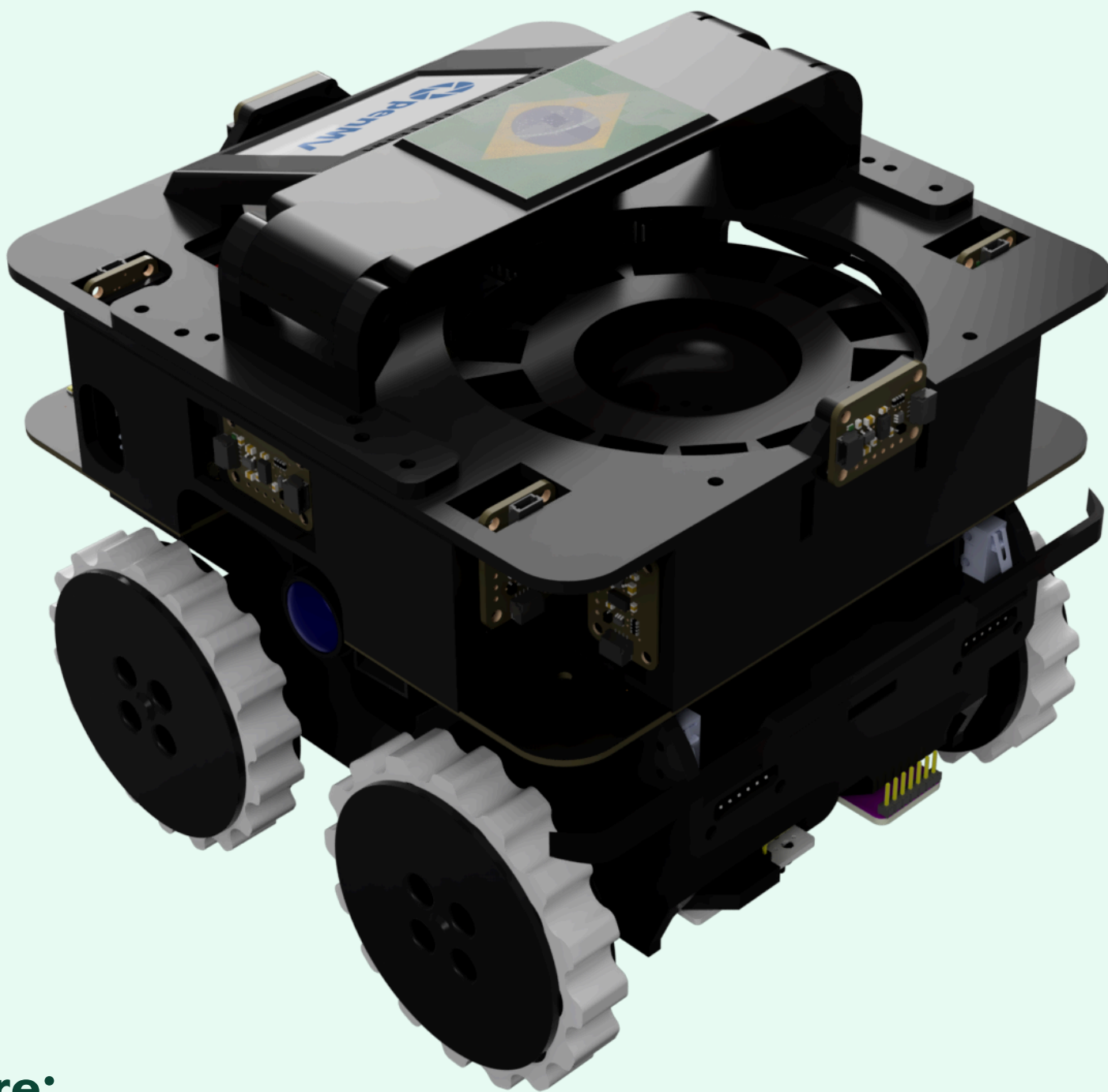


In 2026, for RoboCup, we upgraded our suspension system, introduced new wheel sizes made of silicone, improved our PCB, and replaced the kit system for greater precision.



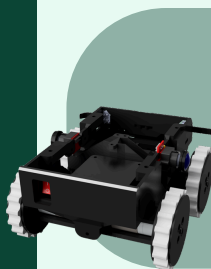
In 2024, we improved our PCB, optimized our kit system — which was too strong and caused the kits to be pushed off the tile — and enhanced our computer vision by implementing CNNs.

In 2025, for CBR and RoboCup, we developed a suspension system for the robot and new wheels made of silicone. Additionally, we improved both our PCB and our kit system.



Hardware:

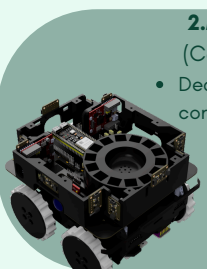
The robot features a three-layer modular design that optimizes functionality and space utilization:



1.Lower Layer (Base Platform):

- Optimized space allocation for upper layers.
- Strategic component placement for electrical efficiency
- Suspension system.

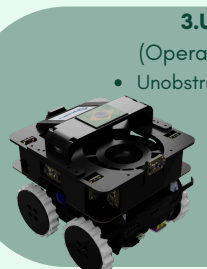
Font : Authors, 2026



2.Middle Layer (Control Center):

- Dedicated space for electrical components
- Efficient cable management
- Protection for sensitive electronics.

Font : Authors, 2026



3.Upper Layer (Operational Interface):

- Unobstructed camera field of view
- System status display interface
- Carrying handle for manual transport

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Electrical Components:

The team's Printed Circuit Board(PCB): The team designed a board, ilustred in the image, specially made thinking the bot functions, to comport our components, reduce the electrical structure, simplify and maximize the functionality also.

Distance Sensor



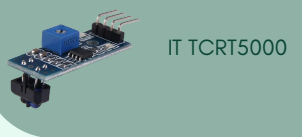
ToF - VL53L4CD

Color Sensor for Blue Tile



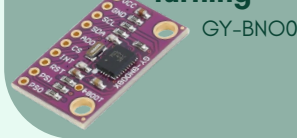
TCS34725

Infrared Sensor for Black tile



IT TCRT5000

Acceleration and Turning



GY-BNO08X

Change I2C Sensor Adress



PCF8574N

Serial Monitor Display



Display Nextion

Realease of the kits



STS3032 - Servo

Motor



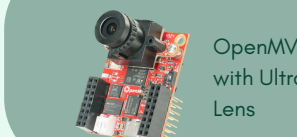
63:1 Metal Gearmotor 20Dx43L mm 12V CB

H-bridge Driver & Controller



M5Stack - 4 Encoder Driver Module

Camera



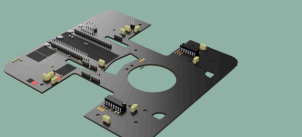
OpenMV RT1062 with Ultrawide Lens

Microcontroler



ESP32 P4

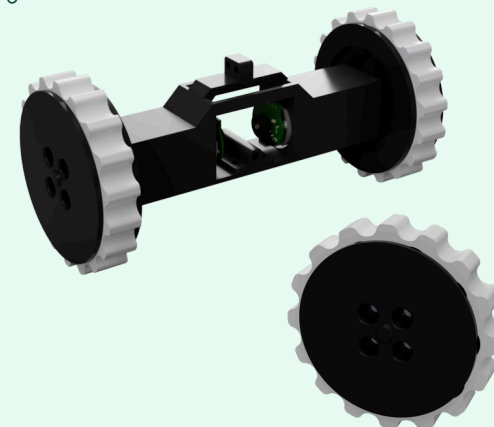
PCB



Innovation:

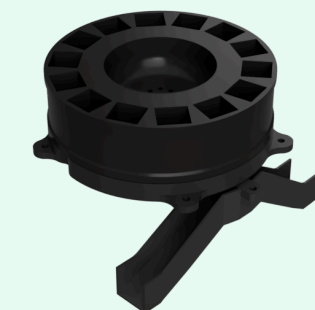
• Suspension and Wheels:

The suspension system consists of two springs and operates as follows: the gearbox applies a contact force to the silicone wheels, compressing the springs and lifting the robot. Made of silicone, the wheels improve grip, and the combined system allows a maximum tilt angle of 13.5 degrees.



• Rescue Kit:

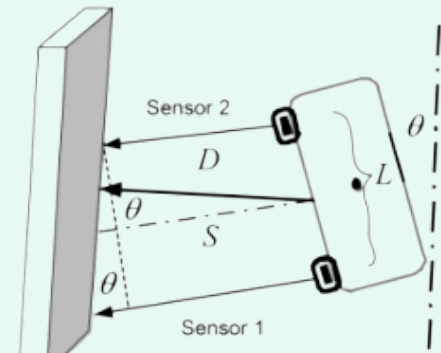
The kit system features a servo motor that selects the side for kit deployment. Strategically positioned ramps guide the kit, ensuring accurate placement within the tile. This design improves reliability and precision during deliveries in competition.



League: RCJ Maze

Software

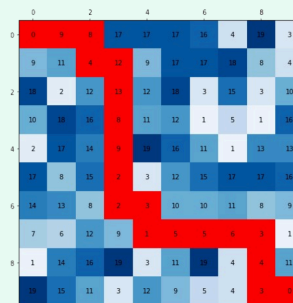
The robot's software system was designed to ensure precise and reliable motion control while navigating the RoboCup 2026 Rescue Maze. To achieve this, four independent PID controllers (Proportional, Integral, and Derivative) were implemented, responsible for maintaining wall alignment, correcting orientation angles, and ensuring directional stability. These controllers operate based on sensor fusion, which includes VL53L4CD time-of-flight sensors for accurate distance measurement, wheel encoders for velocity control, and filtering algorithms such as Kalman filtering and moving average to reduce sensor noise. This combination allows the robot to continuously adapt to environmental conditions, maintaining smooth and accurate movement even in narrow passages or sharp turns.



PID Control Logic Diagram
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Mapping and Navigation

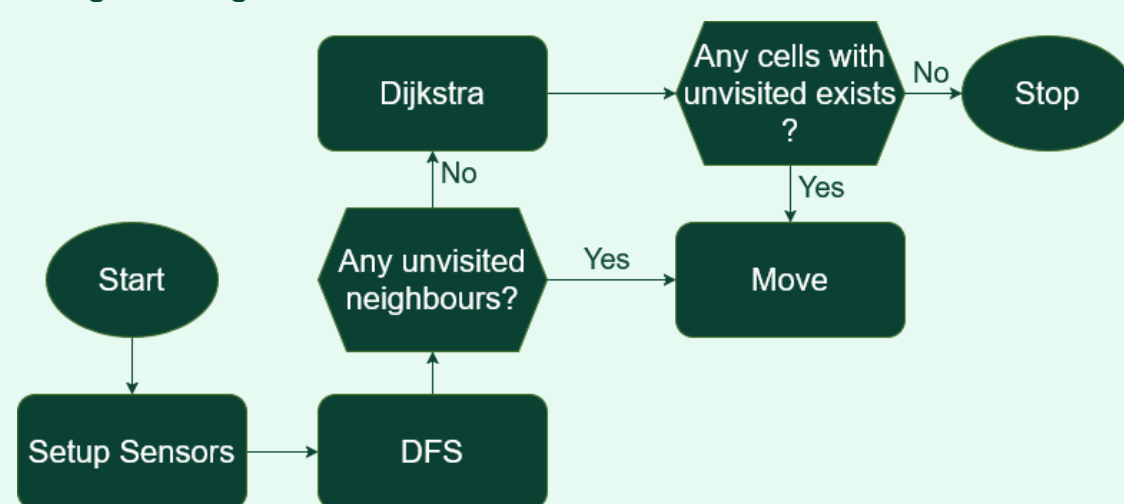
While moving through the maze, the robot performs real-time mapping using a matrix-based representation to record wall locations and explored areas. To determine its path, it employs a hybrid approach combining DFS (Depth-First Search) and Dijkstra algorithms. DFS allows the robot to explore deeply into a path before backtracking, while Dijkstra provides a broader analysis, checking nearby paths and seeking shorter routes. This combined strategy ensures adaptive and complete navigation, enabling the robot to systematically explore the entire maze, avoid revisiting previously explored regions, and return to the starting point using reconstructed paths. As a result, the system effectively handles various maze structures, including branches, dead ends, and complex routes, ensuring efficient and comprehensive exploration.



Dijkstra Maze Exploration Visualization

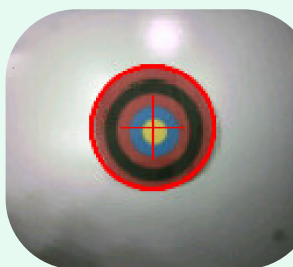
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Navigation Algorithms:



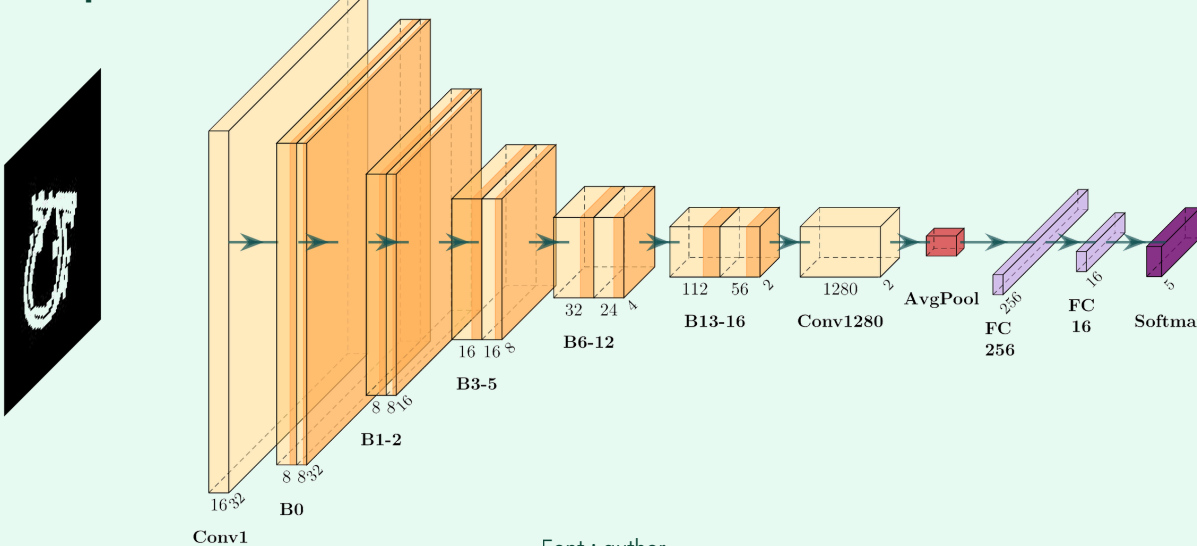
Victim Identification

To identify simulated victims throughout the maze, the robot features an embedded computer vision system based on the OpenMV RT1062 camera. This system is capable of real-time image processing using Convolutional Neural Networks (CNNs) for accurate pattern recognition. The software also includes multi-threshold color detection and circle recognition algorithms, enabling reliable identification of visual targets and Greek letters. The system is optimized for performance, achieving a detection latency of under 100 milliseconds and a classification accuracy of 98.5%, even in environments with variable lighting. This capability ensures the robot can operate autonomously and efficiently, quickly and accurately identifying victims — an essential function in realistic rescue simulations.



Captured through our robot's vision system
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Computer Vision



Font : author

